Checkpoint 1 — ERE Detection Pipeline Progress Summary

1. Scientific Background — Nikumbh et al. (2019)

Goal: Replicate Nikumbh et al. (2019) "Recent spatial aggregation tendency of rainfall extremes over India." Original used IMD gauge data 1951-2015. This project uses ERA5 1998-2020 at 1-degree resolution.

Core finding: ERE average size S-bar has significantly increased since 1980. Frequency NE is flat. Events are getting larger, not more frequent.

Three quantities:

- NT = total extreme grids per day
- NE = number of distinct connected ERE objects per day
- S-bar = NT / NE = average ERE size per day

4-connectivity: Two extreme grids are one ERE only if they share a full edge (N/S/E/W). Diagonal contact = separate events. Default in scipy.ndimage.label().

Loc_ERE stores SIZE not label: Every grid in an ERE stores the count of connected grids. NaN means no ERE that day.

ERE size classification:

Category	Size	Area
Small	1 grid	~12,000 km2
Medium	2-5 grids	~60,000 km2
Large	>=6 grids	>70,000 km2

Relationship between quantities:

Observation	Implication
NT rising, NE flat	S-bar increasing
NT rising, NE rising	More events (frequency increase)
NT flat, S-bar rising	Fewer but larger events

2. Data Acquisition — ERA5 Reanalysis

Unit issue found and fixed: ERA5 precipitation is a running hourly accumulation. Requesting 00:00 only gives 1 hour of rain, not 24 hours. Values were ~24x too small.

Fix: Download all 24 hourly timesteps per year. Sum using ds["tp"].resample(valid_time="1D").sum() * 1000 to get true daily totals in mm/day.

Download strategy: Year by year 1998-2020. Each year: download hourly (~300 MB), aggregate to daily (~0.7 MB), delete hourly file immediately. Checkpoint saved per year. Merged all 23 years into one 17 MB final NetCDF.

Persistent storage: Uploaded to Kaggle Dataset "era5-india-daily-precip". Attached to any future notebook — no re-downloading needed.

Final shape: (2806, 36, 41) — 2806 monsoon days (June-Sept 1998-2020), 36 lat points (5N-40N), 41 lon points (60E-100E).

Before and after unit fix:

Metric	Wrong (1 hr)	Correct (24 hr sum)
Max precip	41.1 mm/day	343.5 mm/day
Mean precip	0.24 mm/day	5.73 mm/day
Central India	0.33 mm/day	5.7 mm/day
Expected	6-8 mm/day	6-8 mm/day

3. ERE Detection Pipeline — 5 Steps

Step 1 — Mask dry days: Copy data. Set values <= 0 to NaN. Percentile computed on rainy days only.

Step 2 — Per-grid 99th percentile: Loop over all 1476 grids. Extract full time series, drop NaNs, compute 99th percentile. Result: threshold map (36x41). Western Ghats ~57 mm/day, Pakistan/desert ~13 mm/day.

Step 3 — Exceedance mask: Broadcast threshold to 3D (2806,36,41). Missing precip -> 0, missing threshold -> 999. Boolean mask c = (precip > threshold). Fraction exceeding: 0.87%.

Step 4 — Connected component labelling: For each of 2806 days: ndimage.label() groups adjacent True cells, bincount counts grids per label, size stamped onto all member grids in Loc_ERE.

Step 5 — Save NetCDF: Esize[time,lat,lon], NT_daily[time], NE_daily[time].

Example Loc_ERE on one day:

```
NaN NaN NaN NaN NaN
NaN 4 4 NaN NaN
NaN 4 4 NaN 2
NaN NaN NaN NaN 2
NaN NaN NaN NaN NaN
```

ERE A = 4 connected grids, stores value 4
ERE B = 2 connected grids, stores value 2
NaN = no ERE present at that location

Key design choice: Size is stamped during grouping so downstream code reads size directly from the array without re-running the algorithm.

Cell 2 (threshold) takes ~2 min — nested loop over 1476 grid cells. Cell 6 (detection) takes ~3 min — loop over 2806 days. All other cells are fast.

4. Results and Validation

Detection statistics (correct ERA5 data):

- Mean EREs per day: 6.19
- Max EREs one day: 23 (2007-06-23)
- Mean NT per day: 9.75, Mean S-bar: ~1.57 grids
- Threshold-rainfall correlation: 0.839

Active vs weak monsoon years (PASS):

- Weak years 2002 2004 2009 2014 2015: mean NT = 11.84
- Active years 2007 2010 2011 2019 2020: mean NT = 17.07
- 2007 highest: NT=20.52, max rain=492.96 mm/day
- 1999 lowest: consistent with El Nino weak monsoon

Spatial pattern (PASS):

- Western Ghats: 28.6 ERE-days over 23 years
- Central India (monsoon trough): 27.4 ERE-days
- Pakistan/desert: 7.4 ERE-days (correctly lowest)

Validation summary:

Test	Result
Known events (4 events)	1/4 detected
Seasonal cycle NT	FAIL — June peaks
Seasonal cycle NE	PASS — July peaks
Active vs weak years	PASS
Spatial pattern	PASS
Threshold sanity	PASS

June NT artefact: June has fewer rainy days historically so its 99th percentile threshold is lower. More grids exceed a lower threshold even without higher actual intensity. Not fixed — Nikumbh paper also uses a single seasonal threshold so changing it would make results incomparable. NE correctly peaks in July confirming the connectivity detection is working.

Known events not detected: ERA5 at 1-degree resolution smooths rainfall over ~12,000 km2. Mumbai 2005 (944 mm at one station) becomes ~72 mm when averaged across the grid cell — below the 107 mm local threshold. This is a dataset limitation, not an algorithm failure. Uttarakhand 2013 was detected (151 mm, threshold 87 mm, ERE size 19 grids).

5. Sullivan et al. (2019) — MCS Tracking Reference Paper

Paper: "The Response of Tropical Organized Convection to El Nino Warming." Tracks Mesoscale Convective Systems using ISCCP satellite IR data at 3-hourly resolution 1983-2008. Same 3-step pipeline: detect, cluster, track. Directly relevant to ERE tracking design.

Their tracking rules:

- Draw 5x5 degree search box around system centre at time t
- Find candidate systems in the next image (+3 hours) within that box
- Require at least 15% areal overlap between system and candidate
- If multiple candidates, assign to the most similar by property difference
- Properties used: location, horizontal extent, cloud top temperature Ttop

Merge: Terminate smaller track, continue larger. Merges are implicit in the database — not recorded as explicit events.

Split: Keep most similar daughter as continuation, initiate second daughter as new track. Also implicit.

Key quantitative finding: Merges far outnumber splits. Expected to hold for Indian monsoon EREs given the organising role of the monsoon trough and low pressure systems.

Adaptations needed for daily ERE tracking:

Aspect	Sullivan MCS	ERE adaptation
Time resolution	3-hourly	Daily — needs larger search radius
Overlap rule	15% areal	Grid cell count ratio (discrete)
Search box	5 deg box	5 deg radius OR direct overlap
Merge handling	Implicit	Explicit recorded event
Split handling	Implicit	Explicit recorded event
Similarity props	Loc+extent+Ttop	Loc+size+rainfall intensity

Overlap on discrete 1-degree grid: Sullivan uses continuous geometry at 30km pixel resolution. On your 1-degree grid, overlap is simply shared grid cells divided by smaller ERE size. No continuous geometry needed — just set intersection.

Split similarity score:

- Primary: centre of mass distance in degrees (location)
- Secondary: abs(parent size - daughter size) / parent size
- Tertiary: abs(parent mean rain - daughter mean rain) / parent
- Daughter with lower total score = continuation of parent track

6. Next Steps — ERE Tracking Algorithm Design

Agreed tracking design decisions:

- Link criterion: >=15% shared grids OR centre within 5 degrees
- Gap allowed: 0 days — consecutive days only
- Season boundary: hard break — no linking across year boundaries
- Output: explicit merge and split event database

Graph-based event classification:

- Build bipartite overlap matrix between EREs on day t and day t+1
- Find connected components of the graph
- 1-to-1 = continuation, 1-to-N = split, N-to-1 = merge
- N-to-N = complex event (merge and split simultaneously)
- 0-to-1 = birth, 1-to-0 = death

Improvements over Sullivan:

- Merges and splits explicitly detected and recorded
- Full metadata per event: date, parent/child sizes and locations
- Complex many-to-many events handled via graph components
- Domain edge events flagged as exit not death

Physical sanity checks for tracking output:

- Merges should outnumber splits (Sullivan et al. finding)
- Merge frequency should peak in July-August (active trough)
- Large ERE merges should concentrate in Central India
- Active years 2007 2010 should show more merges than weak years

Metadata to record per event:

Field	Description
date	Day when event occurs
event_type	merge / split / birth / death
parent_ids	ERE labels on day t
child_ids	ERE labels on day t+1
parent_sizes	Grid cell counts of parents
child_sizes	Grid cell counts of children
location	Centre of mass lat/lon
mean_rain	Mean rainfall intensity mm/day

Checkpoint 1 complete — ERA5 corrected dataset ready on Kaggle, ERE pipeline validated, tracking design agreed. Next: review MCS tracking reference code and implement ERE tracking algorithm.